

A Lean-verified offset construction for the Erdős–Straus equation

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Abstract

We give a self-contained formulation of an offset construction for the Erdős–Straus equation

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

The construction packages the classical divisor split into a congruence condition indexed by a prime $p \equiv 3 \pmod{4}$. Its full positive-integer statement is formalised in Lean 4 with Mathlib: the development constructs the denominators by natural-number division, proves the divisions exact and the denominators positive, and concludes the equality in $\mathbb{Q}$. We also formalise a fixed-divisor lifting theorem, an exact divisibility test for arbitrary transport steps, and three groups of explicit infinite families, including the progressions arising from $p = 3$ and $d_1 = 2, 5$. We make no claim toward resolving the Erdős–Straus conjecture; the underlying divisor split is classical.

1 The divisor split

The Erdős–Straus conjecture asks whether, for every integer $n \geq 2$, there exist $x, y, z \in \mathbb{N}_{>0}$ satisfying

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}. \tag{1}$$

The conjecture remains open. We use the following classical algebraic reduction; for historical background, see Mordell [1] and Vaughan [2].

Fix $x \in \mathbb{N}_{>0}$ with $4x > n$, and set

$$a = 4x - n, \quad b = nx.$$

Then

$$\frac{4}{n} - \frac{1}{x} = \frac{a}{b},$$

and multiplication and rearrangement give

$$(ay - b)(az - b) = b^2. \tag{2}$$

Lemma 1.1 (positive divisor split). *Let $n, x, a, b, d_1, d_2, y, z \in \mathbb{N}_{>0}$. Suppose*

$$a = 4x - n, \quad b = nx, \quad d_1 d_2 = b^2,$$

and

$$ay = b + d_1, \quad az = b + d_2.$$

Then (x, y, z) satisfies (1).

Proof. The last three equations are precisely the factorisation (2). Expanding it and using $a = 4x - n$ and $b = nx$ gives

$$4xyz = n(xy + xz + yz).$$

All four variables n, x, y, z are nonzero, so division by $nx y z$ gives (1). $\square$

The Lean development separates these two steps. It first proves the cross-multiplied identity over $\mathbb{Z}$, then proves that a positive instance over $\mathbb{N}$ gives the rational equality in $\mathbb{Q}$.

2 A positive certificate

The offset form avoids subtraction and is convenient both mathematically and formally.

Theorem 2.1 (positive offset certificate). *Let $n, x, p, b, d_1, d_2, y, z \in \mathbb{N}$, with $n, x, p, y, z > 0$. If*

$$4x = n + p, \quad b = nx, \quad (3)$$

$$d_1 d_2 = b^2, \quad py = b + d_1, \quad pz = b + d_2, \quad (4)$$

then (x, y, z) satisfies (1).

Proof. The equations are the positive divisor split with $a = p = 4x - n$. Equivalently, substitution in the cross-multiplied target leaves the multiple

$$p(d_1 d_2 - b^2),$$

which vanishes by (4). Positivity then permits passage to the rational identity. $\square$

In Lean, `IsES n x y z` contains both the four positivity assertions and the equality (1) in $\mathbb{Q}$. Theorem 2.1 is the declaration `isES_of_offset_certificate`.

3 The constructive offset theorem

Theorem 3.1 (constructive offset theorem). *Let $n, p, d_1 \in \mathbb{N}$ satisfy*

$$n \geq 2, \quad p \text{ prime}, \quad p \equiv 3 \pmod{4}, \quad n \equiv 1 \pmod{4}, \quad p \nmid n.$$

Set

$$x = \frac{n+p}{4}, \quad b = nx.$$

Suppose $d_1 > 0$, $d_1 \mid b$, and

$$p \mid b + d_1, \quad (5)$$

equivalently $d_1 \equiv -b \pmod{p}$. Define

$$d_2 = \frac{b^2}{d_1}, \quad y = \frac{b + d_1}{p}, \quad z = \frac{b + d_2}{p}. \quad (6)$$

Then $x, y, z \in \mathbb{N}_{>0}$, all divisions in (6) are exact, and (x, y, z) satisfies (1).

Proof. The congruences on n and p show that $4 \mid n + p$, so $x \in \mathbb{N}$ and $4x = n + p$. Moreover $x > 0$. If $p \mid x$, then $p \mid 4x = n + p$, hence $p \mid n$, a contradiction. Thus $p \nmid b = nx$.

Since $d_1 \mid b$, the quotient $d_2 = b^2/d_1$ is exact and positive. It remains to prove $p \mid b + d_2$. Work in the field $\mathbb{Z}/p\mathbb{Z}$. Condition (5) gives

$$d_1 = -b.$$

The element b is nonzero, hence so is d_1 . From $d_1 d_2 = b^2$, we may cancel d_1 to obtain

$$d_2 = -b,$$

and therefore $p \mid b + d_2$. Both definitions of y, z in (6) are consequently exact. Their numerators are positive, so $y, z > 0$. Theorem 2.1 now applies. $\square$

Remark. The primality assumption is used only for cancellation modulo p . The Lean proof implements this step in `ZMod p`; the remainder of the construction uses natural numbers, with the final equality stated over the rationals.

4 Fixed-divisor lifting

For fixed p , define

$$X_p(n) = \frac{n+p}{4}, \quad B_p(n) = nX_p(n),$$

where division is natural-number division. Call a triple (p, d, n) *offset-admissible* if

$$\begin{aligned} n \geq 2, \quad p \text{ prime}, \quad p \equiv 3 \pmod{4}, \quad n \equiv 1 \pmod{4}, \quad p \nmid n, \\ d > 0, \quad d \mid B_p(n), \quad p \mid B_p(n) + d. \end{aligned}$$

These are exactly the hypotheses of Theorem 3.1, packaged with a fixed divisor d .

Theorem 4.1 (fixed-divisor lifting). *If (p, d, n) is offset-admissible, then for every $k \in \mathbb{N}$,*

$$n_k = n + 4pdk$$

is offset-admissible for the same p, d . Consequently, on setting

$$x_k = X_p(n_k), \quad b_k = B_p(n_k), \quad d_{2,k} = \frac{b_k^2}{d},$$

the positive natural numbers

$$y_k = \frac{b_k + d}{p}, \quad z_k = \frac{b_k + d_{2,k}}{p}$$

satisfy

$$\frac{4}{n_k} = \frac{1}{x_k} + \frac{1}{y_k} + \frac{1}{z_k}.$$

Moreover, $k \mapsto n_k$ is strictly increasing, so these parameters form an infinite arithmetic progression.

Proof. The congruences defining admissibility make $n + p$ divisible by 4. For $n_k = n + 4pdk$, exact division gives

$$X_p(n_k) = X_p(n) + pdk. \tag{7}$$

Expanding $B_p(n_k) = n_k X_p(n_k)$ using (7) gives the exact identity

$$B_p(n_k) = B_p(n) + pdk(n + 4X_p(n) + 4pdk). \tag{8}$$

The added term in (8) is divisible by both d and p . It follows respectively that $d \mid B_p(n_k)$ and $p \mid B_p(n_k) + d$. Also $n_k \equiv n \pmod{4}$ and $n_k \equiv n \pmod{p}$, so the two modular hypotheses are preserved; all other admissibility conditions are unchanged or immediate from $n_k \geq n$. Theorem 3.1 supplies the displayed positive representation, including exactness of all divisions. Finally,

$$n_{k+1} - n_k = 4pd > 0$$

because a prime p and the admissible divisor d are positive. $\square$

5 An exact test for transport steps

The step $4pd$ in Theorem 4.1 is sufficient, but its proof does not by itself decide whether a smaller step works for a particular admissible seed. The following criterion answers that question without asserting that $4pd$ is always minimal.

Theorem 5.1 (transport-step criterion). *Let (p, d, n) be offset-admissible and let $T > 0$. Put $u = T/4$, using natural-number division. Then*

$$(p, d, n + Tk) \text{ is offset-admissible for every } k \in \mathbb{N}$$

if and only if

$$4 \mid T, \quad p \mid T, \quad d \mid 8u^2, \quad d \mid u(n + 4X_p(n) + 4u). \quad (9)$$

Proof. If the progression remains admissible, its congruence modulo 4, evaluated at $k = 1$, forces $4 \mid T$. Also $p \mid T$: otherwise T is invertible modulo p , and some residue k satisfies $n + Tk \equiv 0 \pmod{p}$, contrary to admissibility.

Write $T = 4u$. Exact expansion gives

$$B_p(n + 4uk) - B_p(n) = u(n + 4X_p(n))k + 4u^2k^2. \quad (10)$$

For natural numbers a_1, a_2, d ,

$$d \mid a_1k + a_2k^2 \quad \text{for every } k \in \mathbb{N}$$

if and only if

$$d \mid a_1 + a_2, \quad d \mid 2a_2.$$

The forward implication follows by evaluating at $k = 1, 2$; the reverse implication uses that $k(k - 1)$ is even. Applying this observation to (10) yields the last two conditions in (9).

Conversely, the four conditions preserve respectively the congruence modulo 4, nondivisibility by p , and the two divisibility clauses involving B_p . The remaining positivity and primality clauses are unchanged, so every transported triple is offset-admissible. $\square$

For example, the criterion proves directly that $T = 12$ does not preserve the seed $(p, d, n) = (3, 2, 5)$, whereas Theorem 4.1 supplies the step $T = 24$. This negative check is included as a Lean regression.

6 Explicit infinite families

We record the families formalised in the accompanying Lean file. In every corollary, $k \in \mathbb{N}$.

Corollary 6.1 (Family I). *Let*

$$n = 12k + 9, \quad x = 3k + 3, \quad b = n(k + 1).$$

Then

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{b+1} + \frac{1}{b(b+1)}.$$

Proof. Here $4x - n = 3$ and $\gcd(3, nx) = 3$, so the reduced numerator is 1. Taking $d_1 = 1$ and $d_2 = b^2$ in the divisor split gives the displayed denominators. The Lean proof is a direct positive polynomial identity. $\square$

Corollary 6.2 ($p = 3, d_1 = 2$). *Let*

$$n = 24k + 5 \quad \text{or} \quad n = 24k + 13.$$

Put $x = (n + 3)/4$ *and* $b = nx$. *Then*

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{(b+2)/3} + \frac{1}{(b+b^2/2)/3}.$$

Proof. The triples $(3, 2, 5)$ and $(3, 2, 13)$ are offset-admissible. Applying Theorem 4.1 gives periods $4 \cdot 3 \cdot 2 = 24$ and hence the two displayed progressions. $\square$

Corollary 6.3 ($p = 3, d_1 = 5$). *Let*

$$n \in \{60k + 5, 60k + 17, 60k + 25, 60k + 37\}.$$

Put $x = (n + 3)/4$ *and* $b = nx$. *Then*

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{(b+5)/3} + \frac{1}{(b+b^2/5)/3}.$$

Proof. The triples with $p = 3, d_1 = 5$ and seeds

$$n \in \{5, 17, 25, 37\}$$

are offset-admissible. Theorem 4.1 gives period $4 \cdot 3 \cdot 5 = 60$, yielding the four displayed progressions. $\square$

The two progressions in Corollary 6.2 are the nonmultiples of 3 in the classical Mordell region $n \equiv 5 \pmod{8}$; the remaining progression $n = 24k + 21$ is contained in Family I. The purpose of including them is to exhibit the general construction and its formal specialisation, not to claim a new residue family.

7 Lean formalisation

The formalisation is contained in `lean/ESTheorem.lean` and uses Lean 4 with Mathlib. The complete source is available at <https://github.com/carlok/erdos-strauss-offset-lean>. Its public positive interface consists of:

- `IsES` and `isES_of_polynomial`;
- `isES_of_offset_certificate`;
- `second_divisibility`;
- `constructive_offset`;
- the reducible definitions `offsetX`, `offsetB`, and the packaged interface `OffsetAdmissible` with `OffsetAdmissible.isES`;
- `offsetX_add_period`, `offsetB_add_period`, and `OffsetAdmissible.add_period`;
- `offsetX_add_general`, `offsetB_add_general`, `poly_dvd_all`, and the exact transport-step criterion `OffsetAdmissible.all_add_mul_iff`;
- `fixed_d_progression` and `fixed_d_progression_strictMono`;
- `family_I_positive`, `family_d2_24k5`, `family_d2_24k13`, and the four `family_d5_60k` declarations.

For every explicit progression, the file also checks the instances $k = 0$ and $k = 1$. The theorem statement includes the positive denominators and the rational unit-fraction equality; the formalisation is therefore not merely a check of a polynomial identity.

8 Scope

The divisor split is classical, and offset choices of x belong to the standard toolkit for Egyptian-fraction problems. The contribution of this note is a compact theorem interface, an end-to-end Lean proof of the positive construction, an exact transport-step criterion, and a small collection of formally checked infinite specialisations. The full Erdős–Straus conjecture remains outside its scope.

References

- [1] L. J. Mordell, *Diophantine Equations*, Academic Press, 1969.
- [2] R. C. Vaughan, *On a problem of Erdős, Straus and Schinzel*, *Mathematika* 17 (1970), 193–198, [doi:10.1112/S0025579300002886](https://doi.org/10.1112/S0025579300002886).
- [3] The Mathlib Community, *Mathlib: the mathematical library of Lean 4*, [project repository](#).